



Pharm*Achieve*

URINARY TRACT INFECTIONS (UTIs)

Qualifying Exam Preparatory Course

COMPETENCIES COVERED IN THIS PRESENTATION...

**Providing Care:
Clinical Care**

**Providing Care:
Distribution**

**Knowledge
& Expertise**

**Communication
& Collaboration**

**Leadership &
Stewardship**

Professionalism

LEGEND

AMG Aminoglycosides

BPH Benign Prostatic Hyperplasia

cUTI Complicated Urinary Tract Infection

CVA tenderness Costovertebral Angle tenderness

ESBL Extended Spectrum B-lactamases

FQ Fluoroquinolones

Hpf High-Power Field

IV Intravenous

N/V Nausea/Vomiting

PEcK *Proteus spp, E. coli, Klebsiella spp*

PO By Mouth

SMX-TMP Sulfamethoxazole/Trimethoprim

STI Sexually Transmitted Infection

UTI Urinary Tract Infection

uUTI Uncomplicated Urinary Tract Infection

DEFINITIONS

Type	Description	Causative pathogens
Asymptomatic Bacteriuria	Bacteriuria in the absence of symptoms	<i>E. coli</i> , <i>P. mirabilis</i> , <i>many others</i>
Uncomplicated UTI	- UTI confined to the bladder (also known as cystitis) in afebrile women or men - Symptoms: dysuria, burning, urgency, frequency, hematuria and suprapubic pain. - No fever or other signs of systemic illness	<i>E. coli</i> >>> <i>P. mirabilis</i> , <i>K. pneumoniae</i> , <i>S. saprophyticus</i> , <i>Enterococcus spp.</i>

Gram negative

Gram positive

UTIs involve microorganisms infecting any part of the urinary tract with the presence of symptoms (not just colonization)

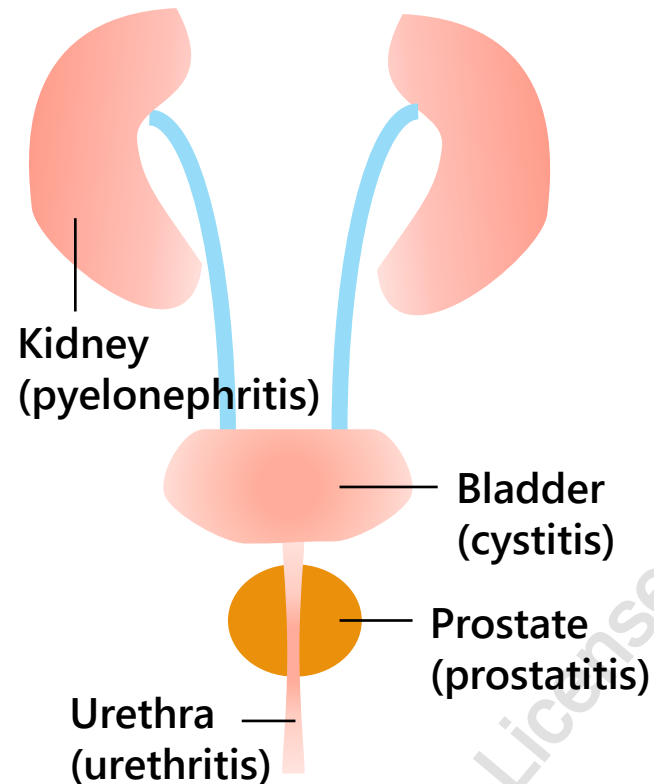
DEFINITIONS

 Gram negative
 Gram positive

Type	Description	Causative pathogens
Complicated UTI	• Infection beyond the bladder in women and men • Includes pyelonephritis, febrile or bacteremic UTI*, catheter-associated UTI, prostatitis** • Symptoms include flank pain, fever, other signs of systemic illness (e.g. chills, hemodynamic instability), costovertebral angle tenderness * Urosepsis: Significant systemic symptoms and/or bacteremia secondary to UTI • Symptoms of sepsis: fever, tachycardia, tachypnea, hypotension ** Prostatitis: Intermittent dysuria, recurrent UTIs, ejaculatory pain, fever, myalgia • Can be asymptomatic	<i>E. coli</i> , > <i>P. mirabilis</i> , <i>Enterococcus faecalis</i> , <i>Staphylococcus</i> , <i>Pseudomonas</i> , <i>Acinetobacter</i>
Recurrent UTI	2 episodes of UTI in the past 6 months or 3 episodes in the past 12 months	Same as initial pathogens (consider antimicrobial resistance)

PATHOPHYSIOLOGY AND RISK FACTORS

Pathophysiology: most common route is ascending colonization of urethra by fecal flora, can ascend to bladder and kidneys



Risk factors



- Age: young children, elderly (BPH, lower estrogen)
- Female gender: short urethra and close proximity to perianal areas
- Pregnancy
- Comorbidities: diabetes, estrogen deficiency
- Urinary tract factors: presence of a catheter, obstruction, renal dysfunction
- Neurologic dysfunction
- Congenital abnormalities
- Previous antibiotic use
- Sexual intercourse, use of diaphragm and spermicides

CAUSES AND PATHOGENS

Pathogens

1. Enterobacteriales

- *Escherichia coli*
- *Klebsiella species*
- *Proteus species* (i.e. *Proteus mirabilis*)

Most common
pathogen ~80%

2. *Staphylococcus saprophyticus* (uncomplicated UTI)

3. *Enterococcus species*

4. Nosocomial pathogens (acquired in hospitalized patients)

- *Pseudomonas species*
- Gram negatives producing extended spectrum β -lactamases (ESBL)



Gram negative



Gram positive

DIAGNOSIS

- Based on symptoms (dysuria, suprapubic tenderness, increased frequency, hematuria) +/- urinalysis/dipstick/urine culture
- Indications for urine culture:
 - **Always** for...
 - Complicated UTI (including pyelonephritis, prostatitis)
 - Pregnant women
 - Diagnostic uncertainty
 - For uncomplicated cystitis, **only** in the following scenarios:
 - Failure to respond to empiric therapy
 - Early recurrence/relapse (< 1 month) following therapy

GOALS OF THERAPY

- ✓ Eradicate the infection and eliminate signs and symptoms
- ✓ Prevent associated complications and recurrence
- ✓ Avoid toxicity associated with selected agent

NON-PHARMACOLOGICAL MEASURES

FYI

Fluid hydration

- Can increase voiding → rapid dilution of bacteria and removal of infected urine
- Residual volume after voiding is critical: even 10 mL can alter eradication
- Fluid intake: 1.5 L/day of water helps prevent recurrent infection

Lactobacillus species probiotics

- ↓vaginal pH, may ↓ colonization of *Escherichia coli*

Cranberry juice/extract

- May reduce rate of recurrence

Surgical procedures

- Surgically correct deformities that predispose the patient to recurrent infections

PHARMACOLOGICAL MEASURES- ANTIBIOTIC THERAPY

Choose based on local resistance data (use antibiogram) and patient factors
No culture required for uUTI unless recurrent/failure

Type of UTI and infecting pathogens	1 st line therapy	2 nd line therapy
Uncomplicated cystitis; single organism • <i>E.coli</i> (80-90%), <i>Klebsiella</i> , <i>Proteus</i> • <i>S.saprophyticus</i> • <i>Enterococcus faecalis</i>	One of the following: • Nitrofurantoin PO x 5 days • SMX-TMP PO x 3 days • Fosfomycin 3 g PO x 1 dose	One of the following: • Fluoroquinolone (ciprofloxacin/levofloxacin/ norfloxacin) PO x 3 days • Beta-lactams x 3-7 days (lower efficacy; avoid ampicillin or amoxicillin alone due to high <i>E. coli</i> resistance)

Type of UTI and infecting pathogens	1 st line therapy
Asymptomatic bacteriuria • <i>E.coli</i> (most common)	• Bacteriuria without symptoms does NOT need to be treated unless the patient is pregnant or undergoing endourological procedures • Target treatment towards sensitivity of organism

EMPIRIC TREATMENT FOR COMPLICATED UTI

Culture all patients!

Choose drug based on local resistance data (use antibiogram) and patient factors
Tailor spectrum and step-down to PO antibiotics once the patient stabilizes

Type of UTI	1 st line therapy options (1 of the following)	Alternatives
With sepsis (with or without shock)	- Fluoroquinolone (levofloxacin, ciprofloxacin) 3rd generation cephalosporins (e.g. ceftriaxone) or 4th generation (cefepime) - Carbapenems - Piperacillin/tazobactam	- AMG (e.g. gentamicin) - Novel beta-lactam-beta lactamase inhibitors (e.g. ceftolozane-tazobactam)
Without sepsis	IV: - Fluoroquinolone (levofloxacin, ciprofloxacin) 3rd generation cephalosporins (e.g. ceftriaxone) or 4th generation (cefepime) - Piperacillin/tazobactam PO: - Fluoroquinolone (levofloxacin, ciprofloxacin) - SMX-TMP	IV: - AMG (e.g. gentamicin) - Novel beta-lactam-beta lactamase inhibitors (e.g. ceftolozane-tazobactam) - Carbapenems PO: - Amoxi-clav - Cephalosporins

Fluid and antipyretic therapy should be administered PRN and improvement should be seen within 72 hours

Nitrofurantoin and fosfomycin are NOT suitable treatment options: they do not achieve adequate levels in the kidneys

Total treatment duration: **5-7 days** if clinical improvement (a longer duration may be warranted in those with delayed response or more severe infection e.g. bacterial prostatitis, indwelling catheters, severe sepsis)

PROSTATITIS TREATMENT

FYI

Type of UTI and infecting pathogens	Symptoms	Treatment
Acute bacterial prostatitis • <i>E.coli</i> • <i>P. aeruginosa</i> • <i>Staph aureus</i> • Others; can be STI-related: <i>N. gonorrhea, C. trachomatis</i>	• Chills, perineal and low back pain, fever, irritative and obstructive voiding • Prostate is tender and swollen	• SMX-TMP or fluoroquinolone • Ceftriaxone then doxycycline if STI related • Do NOT use beta-lactams (unless STI) due to poor penetration • Treat for 4 weeks (in some cases 2 weeks is sufficient)
Chronic bacterial prostatitis • <i>PEcK</i> • <i>P. aeruginosa</i> • Others; can be STI-related: <i>N. gonorrhea, C. trachomatis</i>	• Intermittent urinary infection presenting as cystitis • History of recurrent UTIs • Prostate exam is usually normal	• Same treatment as acute • Treat for 4-6 weeks

RECURRENT URINARY TRACT INFECTIONS TREATMENT

- 2 types:
 - Relapse/persistence: symptomatic bacteriuria with same bacteria as before; can occur during or < 2 weeks after completion of therapy
 - Reinfection: symptomatic bacteriuria with same or different bacteria as before; usually occurs > 2 weeks after completion of therapy
- Recurrent UTI: Defined as ≥ 2 infections in 6 months or ≥ 3 infections in 12 months

Short-course self-therapy

Post-sexual intercourse prophylaxis (post-coital)

- One dose within 2 hours of intercourse
- One of: SMX-TMP, TMP, nitrofurantoin, cephalexin

Long-term low-dose prophylaxis

- One of: SMX-TMP (1st line), TMP, nitrofurantoin, cephalexin, vaginal estrogens and fosfomycin* (*last resort, not used often)
 - Fluoroquinolones (norfloxacin/ciprofloxacin): risk likely outweighs the benefits (e.g. QT prolongation, *C. difficile*)

SPECIAL POPULATIONS

Pregnancy

- Treat asymptomatic bacteriuria to reduce complications
- Treat any symptomatic infection (culture!)
- **Can generally treat similarly to uUTI**
- **Safe:** amoxicillin, amoxicillin/clavulanate, cephalexin, ceftriaxone, fosfomycin
- **Avoid:**
 - Nitrofurantoin in > 36 weeks (1st trimester is controversial; *may* be used in the absence of alternatives)
 - SMX-TMP or TMP in first trimester, SMX-TMP near term (late third trimester)
 - Amoxi-clav in women at risk of preterm delivery
 - Fluoroquinolones, aminoglycosides
- Duration of treatment: 7 days (except for fosfomycin, nitrofurantoin)



Breastfeeding

- **Safe:** fluoroquinolones, nitrofurantoin, SMX-TMP, amoxicillin, amoxicillin/clavulanate, cephalosporins



SPECIAL POPULATIONS

FYI

Children	Catheter-associated UTI	Patients with diabetes and candiduria
• Do not recommend routine screening • UTI should be considered if unexplained fever, hematuria, dysuria • < 2 years old: treat for 7-10 days with antibiotics (TMP, amoxicillin/clavulanic acid, or cephalosporins) • > 2 years old: treat for 3-7 days with antibiotics (TMP, SMX-TMP, amoxicillin/clavulanic acid, or cephalosporins), post-pubescent girls can be treated as adults • FQ are not recommended for use in pediatric and adolescent patients	• Culture, remove catheter (and if needed replace catheter) • Common source of pyelonephritis and bacteremia • Do not treat if patient is asymptomatic • Treat similar to a complicated UTI	• Do not treat asymptomatic • Symptomatic cystitis: choice of antifungal depends on clinical status of the individual patient • Fluconazole is the preferred treatment for Candida UTIs

MONITORING PARAMETERS

Efficacy Endpoints		
Parameter	Degree of Change	Timeframe
Signs and symptoms	Improved and/or Resolved	Improved within hours; resolved within 3 days; if no improvement, then physician reassessment and alternatives may be recommended
Urine culture (if pregnant)	No bacteria	2 weeks later, then monthly until delivery

Safety Endpoints		
Parameter	Degree of Change	Timeframe
Adverse effects from drug and complications	Prevent	Duration of therapy

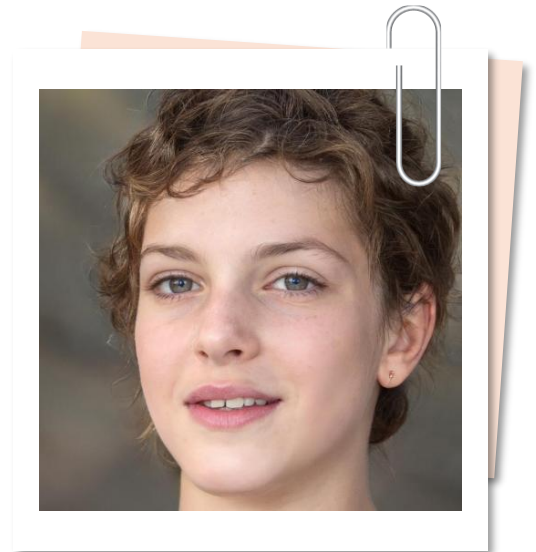
TAKEAWAYS

- UTI is classified by anatomic location and complexity
- *E. coli* is the most common pathogen regardless of site, severity or complexity
- Uncomplicated cystitis is often treated without culture information and diagnosed by history and presentation alone. All other types of UTI should be cultured
- First-line therapy for **uncomplicated UTI**: **nitrofurantoin for 5 days**, **SMX-TMP for 3 days** or **Fosfomycin as a single dose**
- First-line therapy for **complicated UTI**: **fluoroquinolones**, **3rd or 4th generation cephalosporins**, **piperacillin/tazobactam**. May use any of the latter options or **carbapenems** if the patient has sepsis.
- Duration should be defined as total effective antimicrobial therapy with antimicrobials tailored to culture results. A shorter 5 to 7-day duration is preferred if patient is improving clinically.

CASE 1

LY is a 25-year-old female who presents to her GP after noticing blood in her urine. LY began experiencing frequent urination and burning during urination about 3 days ago. LY is also experiencing suprapubic pain. She denies fever, flank pain and vaginal discharge. There has been little/no improvement. LY's dipstick urine test taken today is positive for leukocytes and protein. LY is sexually active with multiple partners, and she takes Yaz® (ethinyl estradiol 0.02 mg and drospirenone 3 mg) 1 tablet PO daily. She has no drug allergies and presents with no fever.

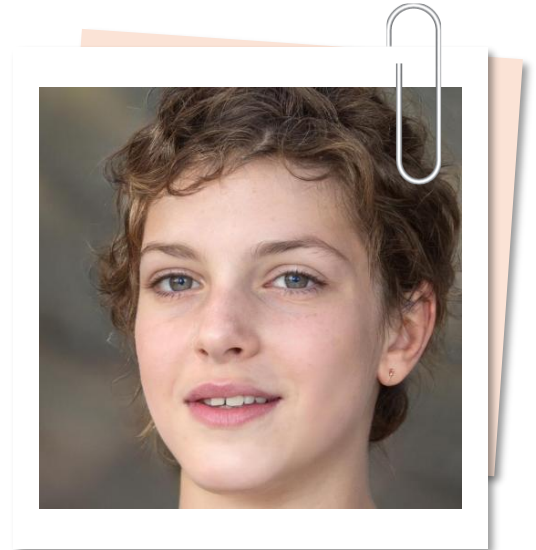
1. What diagnosis most likely explains LY's symptoms?
 - a) Acute uncomplicated cystitis
 - b) Complicated cystitis
 - c) Pyelonephritis
 - d) Gonococcal cervicitis



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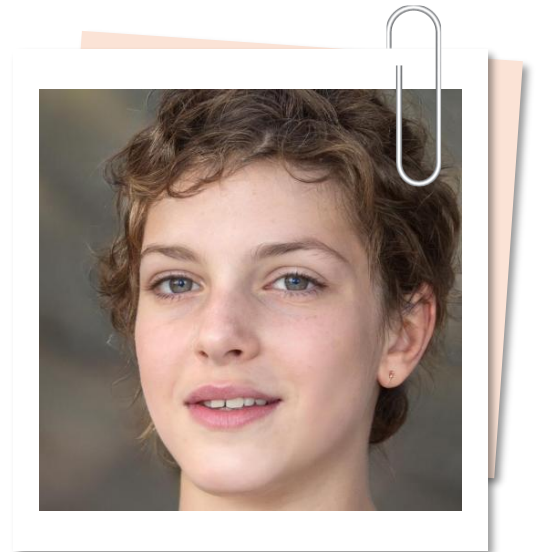
2. Which of the following pathogens is more likely to be causing infection in LY than what would be expected to cause an infection in a 62-year-old female with diabetes?
- a) *P. mirabilis*
 - b) *K. pneumoniae*
 - c) *E. faecalis*
 - d) *S. saprophyticus*



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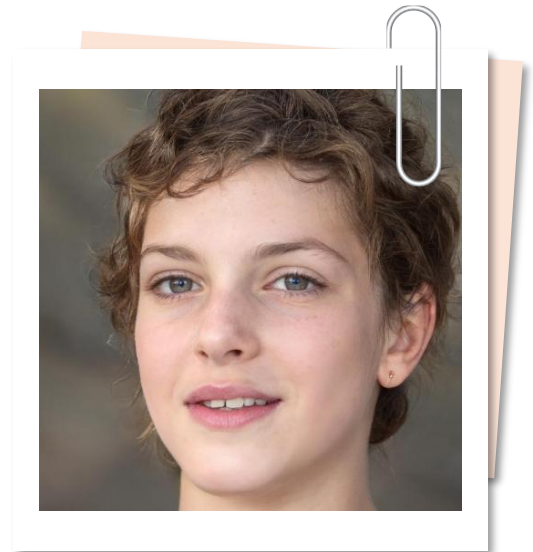
3. What antibiotic regimen is the most appropriate for LY at this time?
- a) Levofloxacin 500 mg PO daily x 3 days
 - b) Nitrofurantoin monohydrate/macrocrystals 100 mg PO BID x 5 days
 - c) TMP 100 mg PO BID x 5 days
 - d) SMX-TMP 800/160 mg PO BID x 5 days



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4. Eight months later, LY experiences these symptoms again and is asking to start prophylaxis. How should LY be managed?
- a) SMX-TMP 800mg/160 mg PO daily
 - b) Cephalexin 500 mg PO BID
 - c) SMX-TMP 800/160 mg after intercourse
 - d) Nitrofurantoin monohydrate/macrocrystals 100 mg PO BID x 5 days



CASE 2

KT is a 68-year-old woman who presents to the emergency department with complaints of a 3-day history of fever, chills, flank pain, and dysuria. She also reports nausea and decreased appetite. The patient mentions that she was experiencing similar symptoms 2 months ago and was prescribed an oral antibiotic but does not recall the name. KT's medical history includes type 2 diabetes (for 7 years) and hypertension (for 5 years). She takes metformin 500 mg BID and ramipril 2.5/hydrochlorothiazide 12.5mg once daily, and her conditions are well controlled. The most recent A1C was 6% (1 month ago). KT is a retired accountant who lives with her partner. She does not smoke but enjoys occasional social alcohol consumption.

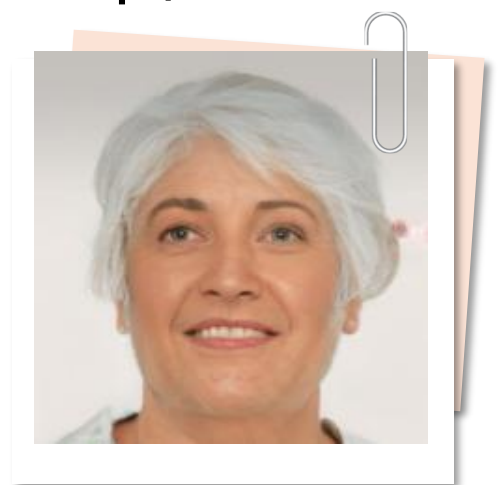
Vitals: temperature 38.7 °C, HR 105 bpm, BP 110/68 mmHg.

Urinalysis: positive for leukocyte esterase, nitrites, and WBC > 20/hpf (normal: 0-5/hpf)

Urine culture: *E. Coli* (pending)

Renal function: eGFR 59 ml/min/1.73 m²

1. Based on the information, how would KT's UTI be classified?
 - a) Uncomplicated cystitis
 - b) Recurrent uncomplicated cystitis
 - c) Pyelonephritis
 - d) Catheter-associated UTI



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Urine culture: *E. Coli* (pending)

Renal function: eGFR 59 ml/min/1.73 m²

2. What is the best empiric antibiotic regimen while awaiting culture results?
 - a) Nitrofurantoin 100 mg PO BID × 5 days
 - b) Amoxicillin 500 mg PO TID × 7 days
 - c) Ciprofloxacin 500 mg Q12H for 7 days
 - d) Moxifloxacin 500 mg once daily for 7 days



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CHANGE LOG

March 2025

- Slide 7: Common pathogen flag moved
- Slide 13: Amox/clav duration changed
- Slide 16: Modified with clarity on drugs in pregnancy
- Slide 24: New references added

June 2025

- Guidelines and slide deck reviewed
- No significant changes: minor edits throughout

October 2025

- Slide 19: Elaborated on Takeaways
- Slides 24-25: Added case 2 and questions

February 2026

- Formatting changes made throughout
- Slide 2: Updated competencies
- Slides 4 and 5: Updated definitions of uncomplicated and complicated UTI based on new IDSA guidelines, and combined definitions for better flow
- Slide 7: Added %incidence of e-coli
- Slide 8: Added pregnant women
- Deleted outdated slides
- Slide 12: New slide on complicated UTI management as per new IDSA guidelines
- Slide 14: Updated info on fluoroquinolone use in recurrent UTI prophylaxis
- Slide 15: Updated info on nitrofurantoin and SMX/TMP in pregnancy, added note about amoxi-clav and preterm delivery (ref 11)
- Slide 16: Updated treatment for catheter-associated UTI
- Slide 26: Updated references